KEVIN O'DONNELL

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EDUCATION

Columbia University New York, NY
***Master of Science in Biomedical Engineering**, Concentration in Robotics and Control* Dec 2024
Coursework: Biomed Innovation, Biostats, Robotics, Control, Mechatronics, Digital Mfg, Lab-To-Market

University of California, Los Angeles (UCLA) Los Angeles, CA
Bachelor of Science in Honors Integrative Biology and Physiology Mar 2022
Awards: Department Honors, Alumni Scholarship National Finalist, Achievement Scholarship

PROFESSIONAL EXPERIENCE

ForUs (Generator Health) Remote
Operations Specialist Feb 2026 - Present
- Automate workflows using Excel/Python to accelerate 100+ daily case resolutions and elevate quality.
- Train team on SOP, coordinate daily workloads, and collaborate with insurers, providers, and patients.

Shape Memory Medical Inc. San Jose, CA
Engineering Intern (cross-functional) Jun 2023 - Aug 2023
- Designed 3 patient-specific cardiac aneurysm models in CAD (SolidWorks), 3D-printed prototypes, and integrated them into a pulsatile flow system for embolization plug R&D Verification & Validation.
- Assisted in device delivery, system calibration, and sensor data collection during preclinical testing.
- Added manual activation control to manufacturing stent crimper circuitry, enhancing system reliability.
- Developed Minitab Design of Experiments to optimize laser welding, increasing tensile strength 367%

Samuel Sia Laboratory - Microfluidics New York, NY
Graduate Research Assistant Aug 2022 - Dec 2022
- Characterized microfluidic pump designs through prototype testing and experimental evaluation, streamlining sample processing and accelerating early COVID detection. Executed PCR testing.

Colwell Laboratory of Circadian and Sleep Medicine Los Angeles, CA
Honors Undergraduate Research Jul 2019 - Jun 2022
- Led a 3-year study/paper on adolescent circadian maturation in wild-type and BLOC-1-deficient mice.
- Developed experimental methods, revealing 5-day faster sleep-wake consolidation in females vs males
- Benchmarked wheel-running against infrared (IR) sensor performance through time-series analysis.

Silk Road Medical Inc. (Acquired by Boston Scientific) Minneapolis, MN
R&D Engineer Intern Jun 2021 - Sep 2021
- Engineered an Arduino carotid artery test bed with sensor components to assess pressure and flow for endovascular catheter/stent procedures, enabling real-time clinician feedback during evaluation.
- Developed post-market surveillance Python tool for root-cause analysis and new product development.

PROJECTS (kevinod.com/Projects)

- Expandable endotracheal intubation model with Design Controls. Interviewed clinical staff and KOLs to translate user needs and risk analysis into design inputs and device requirements. Pitched the project to physicians and investors on a five-person team, won top in class. (CU Biomedical Innovation 2022).
- Python ML tools for healthcare datasets: malaria cell object detection, breast cancer treatment survival modeling, and cardiovascular risk prediction with fairness calibration (>95% AUC-ROC) (2025).
- Robotic pull-wire catheter planning & control algorithm for autonomous vascular navigation (2026).
- Full-stack, API-based medication reminder system for hospital EHR integration (UCLA YTP 2021).
- Robotic Arm with 5-DOF integrating ROS 2, MoveIt 2 path planning, Gazebo/RViz 2 simulation, motion planning, Raspberry Pi Bluetooth real-time control, and electromechanical integration (2025).
- Retinal needle commercialization plan, assessed technical and business viability with CU MBA (2023)

SKILLS

Design Controls, risk analysis, mechatronics, 3D printing, rapid prototyping, DOE, McKinsey Forward
Software: CAD (SolidWorks), Python, C++, machine learning (AI), Minitab, statistics, data analysis